

UNIT IV

CHAPTER 4

Big Data Analytics

Syllabus Topics

Big Data Analytics - Architecture and Life Cycle, Types of analysis, Analytical approaches, Data Analytics with Mathematical manipulations, Data Ingestion from different sources (CSV, JSON, html, Excel, mongoDB, mysql, sqlite); Data cleaning, Handling missing values, data imputation, Data transformation, Data Standardization, handling categorical data with 2 and more categories, statistical and graphical analysis methods, Hive Data Analytics.

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4.1 BIG DATA ANALYTICS LIFE CYCLE IS DIVIDED INTO NINE PHASES, NAMED AS :

UQ. Explain different steps in Data Analytics project life cycle. (SPPU - Q. 5(a), Dec. 19, 8 Marks)

- Phases of Big Data Analytics
- 1. Business Case/Problem Definition
- 2. Data Identification
- 3. Data Acquisition and filtration
- 4. Data Extraction
- 5. Data Munging (Validation and Cleaning)
- 6. Data Aggregation & Representation (Storage)
- 7. Exploratory Data Analysis
- 8. Data Visualization (Preparation for Modeling and Assessment)
- 9. Utilization of analysis results.

Let us discuss each phase :

Phase I : Business Problem Definition

- In this stage, the team learns about the business domain, which presents the motivation and goals for carrying out the analysis.
- In this stage, the problem is identified, and assumptions are made that how much potential gain a company will make after carrying out the analysis.
- Important activities in this step include framing the business problem as an analytics challenge that can be addressed in subsequent phases.
- It helps the decision-makers understand the business resources that will be required to be utilized thereby determining the underlying budget required to carry out the project.
- Moreover, it can be determined, whether the problem identified, is a Big Data problem or not, based on the business requirements in the business case.
- To qualify as a big data problem, the business case should be directly related to one(or more) of the characteristics of volume, velocity, or variety.

Phase II : Data Definition

- Once the business case is identified, now it's time to find the appropriate datasets to work with. In this stage, analysis is done to see what other companies have done for a similar case.
- Depending on the business case and the scope of analysis of the project being addressed, the sources of datasets can be either external or internal to the company.
- In the case of internal datasets, the datasets can include data collected from internal sources, such as feedback forms, from existing software, On the other hand, for external datasets, the list includes datasets from third-party providers.

Phase III : Data Acquisition and filtration

- Once the source of data is identified, now it is time to gather the data from such sources. This kind of data is mostly unstructured. Then it is subjected to filtration, such as removal of the corrupt data or irrelevant data, which is of no scope to the analysis objective. Here corrupt data means data that may have missing records, or the ones, which include incompatible data types.
- After filtration, a copy of the filtered data is stored and compressed, as it can be of use in the future, for some other analysis.

Phase IV : Data Extraction

- Now the data is filtered, but there might be a possibility that some of the entries of the data might be incompatible, to rectify this issue, a separate phase is created, known as the data extraction phase.
- In this phase, the data, which don't match with the underlying scope of the analysis, are extracted and transformed in such a form.

Phase V : Data Munging

- As mentioned in phase III, the data is collected from various sources, which results in the data being unstructured. There might be a possibility, that the data might have constraints, that are unsuitable, which can lead to false results. Hence there is a need to clean and validate the data.
- It includes removing any invalid data and establishing complex validation rules. There are many ways to validate and clean the data.

- For example, a dataset might contain few rows, with null entries. If a similar dataset is present, then those entries are copied from that dataset, else those rows are dropped.

Phase VI : Data Aggregation & Representation

- The data is cleansed and validates, against certain rules set by the enterprise. But the data might be spread across multiple datasets, and it is not advisable to work with multiple datasets. Hence, the datasets are joined together.
- For example: If there are two datasets, namely that of a Student Academic section and Student Personal Details section, then both can be joined together via common fields, i.e. roll number.
- This phase calls for intensive operation since the amount of data can be very large. Automation can be brought into consideration, so that these things are executed, without any human intervention.

Phase VII : Exploratory Data Analysis

- Here comes the actual step, the analysis task. Depending on the nature of the big data problem, analysis is carried out.
- Data analysis can be classified as Confirmatory analysis and Exploratory analysis. In confirmatory analysis, the cause of a phenomenon is analyzed before. The assumption is called the hypothesis.
- The data is analyzed to approve or disapprove the hypothesis. This kind of analysis provides definitive answers to some specific questions and confirms whether an assumption was true or not
- In an exploratory analysis, the data is explored to obtain information, why a phenomenon occurred. This type of analysis answers "why" a phenomenon occurred. This kind of analysis doesn't provide definitive, meanwhile, it provides discovery of patterns.

Phase VIII : Data Visualization

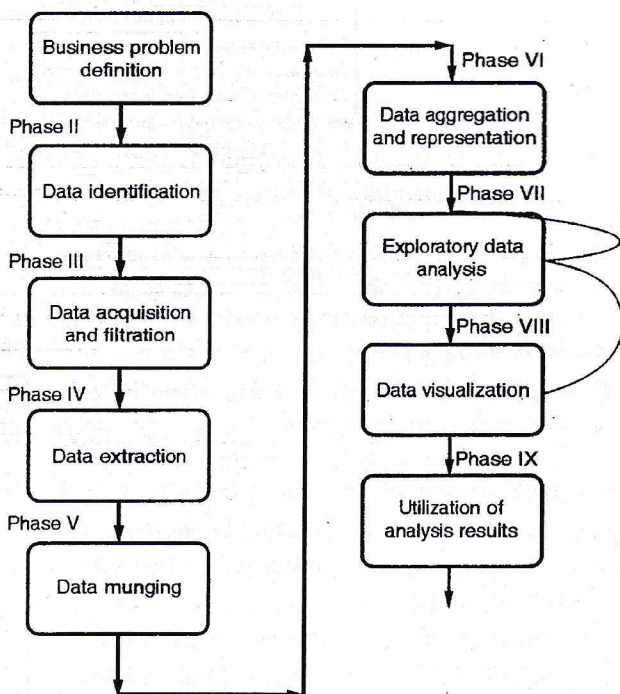
- Now we have the answer to some questions, using the information from the data in the datasets. But these answers are still in a form that can't be presented to business users.

- A sort of representation is required to obtain value or some conclusion from the analysis. Hence, various tools are used to visualize the data in graphic form, which can easily be interpreted by business users. Visualization is said to influence the interpretation of the results.
- Moreover, it allows the users to discover answers to questions that are yet to be formulated.

Phase IX : Utilization of analysis results

- The analysis is done, the results are visualized, now it's time for the business users to make decisions to utilize the results.
- The results can be used for optimization, to refine the business process. It can also be used as an input for the systems to enhance performance.
- The block diagram of the life cycle is given below

Phase I



(101) Fig. 4.1.1 : Life cycle of Big data analytics

It is evident from the block diagram that Phase VII, i.e. exploratory Data analysis, is modified successively until it is performed satisfactorily. Emphasis is put on error correction. Moreover, one can move back from Phase VIII to Phase VII, if a satisfactory result is not achieved. In this manner, it is ensured that the data is analyzed properly.

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4.2 TYPES OF BIG DATA ANALYTICS

UQ. Explain different types of analysis in detail with example.

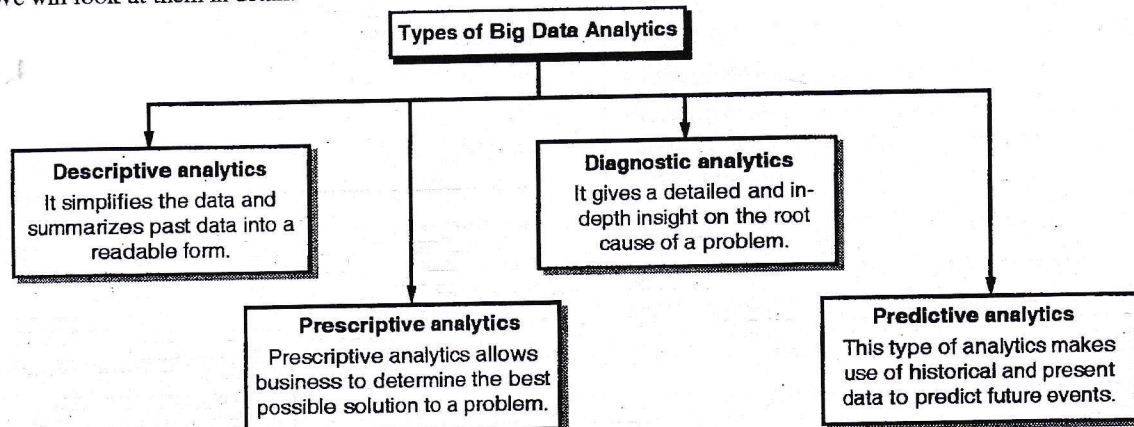
(SPPU - Q. 6(a), Dec. 18, 9 Marks)

The different types of data require different approaches. This different approach of analytics gives rise to the four different types of Big data analytics.

Big data analytics is categorized into four subcategories that are :

1. Descriptive Analytics
2. Diagnostic Analytics
3. Predictive Analytics
4. Prescriptive Analytics

We will look at them in detail.



(102)Fig. 4.2.1 : Types of Big Data Analytics

1. Descriptive Analytics

- Descriptive Analytics is considered a useful technique for uncovering patterns within a certain segment of customers. It simplifies the data and summarizes past data into a readable form.
- Descriptive analytics** provide insights into what has occurred in the past and with the trends to dig into for more detail. This helps in creating reports like a company's revenue, profits, sales, and so on.
- Examples of descriptive analytics include summary statistics, clustering, and association rules used in market basket analysis.
- An example of the use of descriptive analytics is the Dow Chemical Company. The company utilized its past data to increase its facility utilization across its offices and labs.

2. Diagnostic Analytics

- Diagnostic Analytics, as the name suggests, gives a diagnosis to a problem. It gives a detailed and in-depth insight into the root cause of a problem.
- Data scientists turn to this analytics craving for the reason behind a particular happening. Techniques like

drill-down, data mining, and data recovery, churn reason analysis, and customer health score analysis are all examples of diagnostic analytics.

- In business terms, diagnostic analytics is useful when you are researching the reasons leading churn indicators and usage trends among your most loyal customers.
- A use case for diagnostic analytics can be an **e-commerce** company. Given the situation that the sales of the company have gone down even though customers are adding products to their carts.
- The possible reasons behind this problem can be: the form didn't load correctly, the shipping charges are high, and not enough payment methods are available.
- Taking the help of diagnostic analytics, the company comes out with a specific reason and then works on that to resolve the issue.

3. Predictive Analytics

- Predictive Analytics, as can be discerned from the name itself, is concerned with predicting future incidents. These future incidents can be market trends, consumer trends, and many such market-related events.

- This type of analytics makes use of historical and present data to predict future events. This is the most commonly used form of analytics among businesses.
- Predictive analytics doesn't only work for the service providers but also for the consumers. It keeps track of our past activities and based on them, predicts what we may do next.

"The purpose of predictive analytics is NOT to tell you what will happen in the future. It cannot do that. In fact, no analytics can do that. Predictive analytics can only forecast what might happen in the future, because all predictive analytics are probabilistic in nature."

- **Dr. Michael Wu**, Chief AI Strategist, PROS
Predictive analytics uses models like data mining, AI, and machine learning to analyze current data and forecast what might happen in specific scenarios.
- Examples of Predictive analytics include next best offers, churn risk, and renewal risk analysis.

► 4. Prescriptive Analytics

- Prescriptive analytics is the most valuable yet underused form of analytics. It is the next step in predictive analytics.
- The prescriptive analysis explores several possible actions and suggests actions depending on the results of descriptive and predictive analytics of a given dataset.
- Prescriptive analytics is a combination of data and various business rules. The data of prescriptive analytics can be both internal (organizational inputs) and external (social media insights).
- Prescriptive analytics allows businesses to determine the best possible solution to a problem. When combined with predictive analytics, it adds the benefit of manipulating a future occurrence like mitigate future risk.
- Examples of prescriptive analytics for customer retention is the next best action and next best offer analysis.
- A use case of prescriptive analytics can be the Aurora Health Care system. It saved \$6 million by reducing the readmission rates by 10%.
- Prescriptive analytics has good use in the healthcare industry. It can be used to enhance the process of drug development, finding the right patients for clinical trials, etc.

► 4.3 ANALYTICAL APPROACHES

1. Data fusion and data integration

By combining a set of techniques that analyse and integrate data from multiple sources and solutions, the insights are more efficient and potentially more accurate than if developed through a single source of data.

2. Data mining

- A common tool used within big data analytics, data mining extracts patterns from large data sets by combining methods from statistics and machine learning, within database management.
- An example would be when customer data is mined to determine which segments are most likely to react to an offer.

3. Machine learning

Well known within the field of artificial intelligence, machine learning is also used for data analysis. Emerging from computer science, it works with computer algorithms to produce assumptions based on data. It provides predictions that would be impossible for human analysts.

4. Natural language processing (NLP).

Known as a subspecialty of computer science, artificial intelligence, and linguistics, this data analysis tool uses algorithms to analyse human (natural) language.

5. Statistics

- This technique works to collect, organise, and interpret data, within surveys and experiments.
- Other data analysis techniques include spatial analysis, predictive modelling, association rule learning, network analysis and many, many more.
- The technologies that process, manage, and analyse this data are of an entirely different and expansive field, that similarly evolves and develops over time. Techniques and technologies aside, any form or size of data is valuable. Managed accurately and effectively, it can reveal a host of business, product, and market insights.

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4.4 DATA ANALYTICS WITH MATHEMATICAL MANIPULATIONS

What is Data Manipulation?

Data Manipulation Meaning: Manipulation of data is the process of manipulating or changing information to make it more organized and readable. We use DML to accomplish this. What is meant by DML? Well, it stands for Data Manipulation Language or a programming language capable of adding, removing, and altering databases, i.e. changing the information to something that we can read. We can clean and map the data thanks to DML to make it digestible for expression.

Data Manipulation Examples

- Data Manipulation is the modification of information to make it easier to read or more structured. For example, in alphabetical order, a log of data may be sorted, making it easier to find individual entries. On web server logs, data manipulation is also used to allow the website owner to monitor their most famous pages and their sources of traffic.
- Accounting users or related fields also manipulate information to assess the expense of the product, pricing patterns, or future tax obligations. To forecast developments in the stock market and how stocks might perform shortly, stock market analysts also use data manipulation.
- Computers can also use data manipulation to view the information in a more realistic way to users based on code in a user-defined software program, web page, or data formatting.

Purpose of Data Manipulation

For business operations and optimization, data manipulation is a key feature. You have to be able to deal with the data in the way you need it to use data properly and turn it into valuable information such as analyzing financial data, consumer behavior, and doing trend analysis. As such, data manipulation provides an organization with many advantages, including :

- **Consistent data:** It can be structured, read, and better understood by providing data in a consistent format. You may not have a unified view when taking data from various sources, but with data manipulation and commands, you can make sure that the data is structured and stored consistently.
- **Project data:** it is paramount for organizations to be able to use historical data to project the future and to provide more in-depth analysis, especially when it comes to finances. Manipulation of data makes it possible for this purpose.
- **Overall,** being able to convert, update, delete, and incorporate data into a database means you can do more with the data. Create more value from the data. It becomes pointless by providing data that remains static. But you will have straightforward insights to make better business decisions when you know how to use data to your advantage.
- **Delete or neglect redundant data:** data that is unusable is always present and can interfere with what matters.

(A) Contrasted with language programming

It looks very stilted when you first look at Data Manipulation Language. For example, explaining to others how to use a built-in feature in Access is relatively straightforward compared to, using DML to Pick * FROM. DML, however, is not a language for programming. That a machine understands and operates as an implicit program cannot be compiled, or translated into 0s and 1s. Think of it instead as a rather sophisticated formula, as one might find in a spreadsheet. You probably use some very convoluted formulas when using a spreadsheet – DML is simply formula speaking, but for using a database.

Steps Involved in Data Manipulation

When you want to get started with data manipulation, here are the steps you should take into consideration :

1. Only if you have data to do so is data manipulation feasible. You need a database, therefore, which is generated from data sources.
2. This knowledge requires reorganization and restructuring. Manipulation of data helps you to cleanse your information.
3. Import a database and create it for you to work on.

4. You can combine, erase, or merge information through data manipulation.
5. When you manipulate data, data analysis becomes simple.

☛ In Excel, How Do You Manipulate Data?

Manipulation of data in Python and manipulation of data in R are critical aspects of data manipulation. Before moving through the more profound principles of Data Manipulation in Python and R, let us now understand how to manipulate data.

• Most definitely, you are aware of how to use MS Excel. Here are some tips to help you manipulate Excel info.

1. **Formulas and functions** : Addition, subtraction, multiplication, and division are some of the basic math functions in Excel. You need to know how to use these Excel-critical features.
2. **Autofill in Excel** : When you want to use the same equation across several cells, this feature is useful. One way of doing it is to retype the formula. Another way is to drag the cursor to the cell's lower right corner and then downwards. It will help you simultaneously apply the same formula to several rows.
3. **Sort and Filter** : Users can save a lot of time when analyzing data by sorting and filtering options in Excel.
4. **Removing duplicates** : There are often chances of replication of data in the process of collecting and assimilating data. In Excel, the Delete Duplicate feature can help remove duplicate spreadsheet entries.
5. **Column splitting, merging, and merging** : Columns or rows in Excel may often be added or removed. Data organization often requires integrating, splitting, or combining multiple datasets

» 4.5 DATA INGESTION FROM DIFFERENT SOURCES (CSV, JSON, HTML, EXCEL, MONGODB, MYSQL, SQLITE)

☛ Data Ingestion?

Data Ingestion is the process of streaming-in massive amounts of data in our system, from several different external sources, for running analytics & other operations required by the business. Data is ingested to understand &

make sense of such massive amount of data to grow the business.

☛ Where Does this Massive Amount of Data Come From?

- The data is primarily user-generated, generated from IoT devices, social networks, user events are recorded continually which helps the systems evolve resulting in better user experience.
- There is no limit to the rate of data creation. With passing time, the rate grows exponentially. As more users use our app, or IoT device or the product which our business offers, the data keeps growing.
- *Data ingestion is just one part of a much bigger data processing system. More commonly known as handling the Big Data.* The data moves through a data pipeline across several different stages.
- Also, there are several different layers involved in the entire big data processing setup such as the data collection layer, data query layer, data processing, data visualization, data storage & the data security layer.

☛ Different Ways of Ingesting Data

- Data ingestion can be done either in real-time or in batches at regular intervals. It entirely depends on the requirement of our business.
- Data Ingestion in real-time is typically preferred in systems reading medical data like a heartbeat, blood pressure IoT sensors where time is of critical importance. In systems handling financial data like stock market events. These are a few instances where time, lives & money are closely linked.
- On the contrary in systems which read trends over time. For instance, estimating the popularity of the sport over a period of time, we can surely ingest data in batches.

☛ Importance of Data Ingestion

- Businesses today are relying on data. They need user data to make future plans & projections. They need to understand the user needs, his behaviours. All these things enable companies create better products, make smarter decisions, run ad campaigns, give user recommendations, gain a better insight into the market. In short, creating value from data which eventually results in more customer-centric products & increased customer loyalty.

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- There are also other uses of data ingestion such as tracking the service efficiency, getting **everything is okay** signal from the IoT devices used by millions of customers.
- Centralizing records of data streaming in from several different sources like for scanning logs. Reducing the complexity of tracking the system as a whole. Scanning logs at one place with tools like Kibana cuts down the hassle by notches. I'll talk about the data ingestion tools up ahead in the article.

Real-World Industry & Architectural Use Cases of Data Ingestion

Here are some of the use-cases where data ingestion is required.

(i) Moving Massive Amount of Big Data into Hadoop

This is the primary & the most obvious use case. As discussed above, Big Data from all the IoT devices, social apps & everywhere, is streamed through data pipelines, moves into the most popular distributed data processing framework Hadoop for analysis & stuff.

(ii) Moving Data from Databases to Elastic Search Server

- In the past, with a few of my friends, I wrote a product search software as a service solution from scratch with Java, Spring Boot, Elastic Search. Speaking of its design the massive amount of product data from legacy storage solutions of the organization was streamed, indexed & stored to Elastic Search Server. The streaming process is more technically called the Rivering of data.
- As in, drawing an analogy from how the water flows through a river, here the data moved through a data pipeline from legacy systems & got ingested into the elastic search server enabled by a plugin specifically written to execute the task.

(iii) Log Processing, Running Log Analytics Systems

- If your project isn't a hobby project, chances are it's running on a cluster. Monolithic systems are a thing of the past. With so many microservices running concurrently. There is a massive number of logs which is generated over a period of time. And logs are the

only way to move back in time, track errors & study the behaviour of the system.

- Now, when we have to study the behaviour of the system as a whole comprehensively, we have to stream all the logs to a central place. Ingest logs to a central server to run analytics on it with the help of solutions like ELK stack etc.

(iv) Stream Processing Engines for Real-Time Events

- Quick real-time streaming & data processing is key in systems handling LIVE information such as sports. It's imperative that the architectural setup in place is efficient enough to ingest data, analyse it. Figure out behaviour in real time & quickly push information to the fans. After all, the whole business depends on it.
- Let's talk about some of the challenges the development teams have to face while ingesting data.

Challenges Companies Face When Ingesting Data

(i) Slow Process

- Guys, data ingestion is a slow process. How? I'll explain. When data is streamed from several different sources into the system, data coming from each & every different source has a different format, different syntax, attached metadata. The data as a whole is heterogeneous. It has to be transformed into a common format like JSON or something to be understood by the analytics system.
- The conversion of data is a tedious process. It takes a lot of computing resources & time. Flowing data has to be staged at several stages in the pipeline, processed & then moved ahead. Also, at each & every stage data has to be authenticated & verified to meet the organization's security standards. With the traditional data cleansing processes, it takes weeks if not months to get useful information on hand. Traditional data ingestion systems like ETL ain't that effective anymore.

(ii) Complex & Expensive

- As already stated the entire data flow process is resource-intensive. A lot of heavy lifting has to be done to prepare the data before being ingested into the system. Also, it isn't a side process, an entire dedicated team is required to pull off something like that.

- There are always scenarios where the tools & frameworks available in the market fail to serve your custom needs & you are left with no option than to write a custom solution from the ground up.
- The semantics of the data coming from external sources changes sometimes which then requires a change in the backend data processing code too. The external IOT devices are evolving at a quick speed.
- So, these are the factors we have to keep in mind when setting up a data processing & analytics system.

(iii) Data is Vulnerable

When data is moved around it opens up the possibility of a breach. Moving data is vulnerable. It goes through several different staging areas & the development team has to put in additional resources to ensure their system meets the security standards at all times.

Data Ingestion Architecture

- Data ingestion is the initial & the toughest part of the entire data processing architecture.
- The key parameters which are to be considered when designing a data ingestion solution are:
- **Data Velocity, size & format** : Data streams in through several different sources into the system at different speeds & size. Data streams from social networks, IoT devices, machines & what not. And every stream of data streaming in has different semantics. A stream might be structured, unstructured or semi-structured.
- **The frequency of data streaming** : Data can be streamed in continually in real-time or at regular batches. We would need weather data to stream in continually. On the other hand, to study trends social media data can be streamed in at regular intervals.

4.6 DATA CLEANING

- Data cleaning is the process of preparing data for analysis by removing or modifying data that is incorrect, incomplete, irrelevant, duplicated, or improperly formatted.

- This data is usually not necessary or helpful when it comes to analyzing data because it may hinder the process or provide inaccurate results. There are several methods for cleaning data depending on how it is stored along with the answers being sought.
- Data cleaning is not simply about erasing information to make space for new data, but rather finding a way to maximize a data set's accuracy without necessarily deleting information.
- For one, data cleaning includes more actions than removing data, such as fixing spelling and syntax errors, standardizing data sets, and correcting mistakes such as empty fields, missing codes, and identifying duplicate data points.
- Data cleaning is considered a foundational element of the **data science basics**, as it plays an important role in the analytical process and uncovering reliable answers.
- Most importantly, the goal of data cleaning is to create data sets that are standardized and uniform to allow business intelligence and **data analytics tools** to easily access and find the right data for each query.

Uses of Data Cleaning

- Regardless of the type of analysis or **data visualizations** you need, data cleaning is a vital step to ensure that the answers you generate are accurate. When collecting data from several streams and with manual input from users, information can carry mistakes, be incorrectly inputted, or have gaps.
- Data cleaning helps ensure that information always matches the correct fields while making it easier for **business intelligence tools** to interact with data sets to find information more efficiently. One of the most common data cleaning examples is its application in data warehouses.

4.7 HANDLING MISSING VALUES

- UQ.** How Missing values and categorical variables are preprocessed before building model? Explain with example.

(SPPU - Q. 5(b), Dec. 18, 9 Marks)

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Why do missing values occur in data?

Missing values can occur in data for a number of reasons, such as survey non-responses or errors in data entry.

While it may seem that a missing value is a missing value, not all missing data is the same.

Missing data is grouped into three broad categories:

- Missing completely at random.
- Missing at random.
- Missing not at random.

Missing completely at random (MCAR)

Data is missing completely at random if all observations have the same likelihood of being missing.

Some hypothetical examples of MCAR data include :

- Electronic time observations are missing, independent of what lane a swimmer is in.
- A scale is equally likely to produce missing values when placed on a soft surface or a hard surface (Van Buren, 2018).

Missing at random (MAR)

When data is missing at random (MAR) the likelihood that a data point is missing is not related to the missing data but may be related to other observed data.

Some hypothetical examples of MAR data include:

- A certain swimming lane is more likely to have missing electronic time observations but the missing data isn't directly related to the actual time.
- A scale produces more missing values when placed on a soft surface than a hard surface (Van Buren, 2018), independent of the weight.
- Childhood health assessment data is more likely to be missing in lower median income counties.

Missing not at random (MNAR)

When data is missing not at random (MNAR) the likelihood of a missing observation is related to its values. It can be difficult to identify MNAR data because the values of missing data are unobserved. This can result in distorted data.

Some hypothetical examples of MNAR data include :

- When surveyed people with more income are less likely to report their incomes.
- On a health survey, illicit drug users are less likely to respond to a question about illicit drug use.
- Individuals surveyed about their age are more likely to leave the age question blank when they are older.

Type of missing value	Description	Examples	Acceptable solutions
Missing completely at random	All observations have the same likelihood of being missing	• Electronic time observations are missing, independent of what lane a swimmer is in. • A scale is equally likely to produce missing values when placed on a soft surface or a hard surface (Van Buren, 2018). • Geographical location data is equally likely to be missing for all locations.	Deletion, Imputation

Type of missing value	Description	Examples	Acceptable solutions
Missing at random	Likelihood that a data point is missing is not related to the missing data but may be related to other observed data.	- A certain swimming lane is more likely to have missing electronic time observations. - A scale produces more missing values when placed on a soft surface than a hard surface (Van Buren, 2018). - Childhood health assessment data is more likely to be missing in lower median income countries	Deletion, Imputation
Missing not at random	Likelihood of a missing observation is related to its values.	- When surveyed people with more income are less likely to report their income. - On a health survey illicit drug users are less likely to respond to a question about illicit drug use. - Individuals surveyed about their age are more likely to leave the age question blank when they are older.	Imputation

Dealing with missing values

How we should deal with missing data depends both on the cause of the missing values and the characteristics of the data set. For example, we cannot deal with missing **categorical data** in the same manner that we deal with missing **time series data**.

Deletion

When data is MCAR and MAR deletion may be a suitable method for dealing with missing values. However, when data is MNAR, deletion of missing observations can lead to bias.

In this section we cover three methods of data deletion for missing values :

- (A) Listwise deletion (B) Pairwise deletion
(C) Variable deletion

Table 4.7.1 : Methods of deletion for missing values

Method	Description	Advantages	Disadvantages
Listwise deletion	Delete all observations where the missing values occur.	Easy to implement.	- Can result in biased estimates if the missing values are not MCAR. - Wastes useful information. - Can disrupt time series analysis by creating gaps in dates used for analysis.
Pairwise deletion	Uses all available data when computing means and covariances.	- Simple to implement. - Uses all available information.	- Can result in biased estimates if the missing values are not MCAR. - Results in different sample sizes being used for different computations. - Requires that data follow a normal distribution
Variable Deletion	Eliminate a variable from analysis if it is missing a large percentage of observations.	Easy to implement.	- Significant loss in information. - May result in missing variable bias.

Example

Consider a small sample of data from the Nelson-Plosser macroeconomic dataset :

year	cpi	gnp	ip	Emp	cpi
1906	-	9.8	33749	28	
1907	-	10	34371	29	
1908	-	-	33246	28	
1909	116.8	8.5	35072	28	
1910	120.1	10	35762	28	

► **A. Listwise deletion**

- Uses only the observations from 1909 and 1910 for all parts of analysis.
- It eliminates the all data in 1906-1908 because of the missing values in gnp and ip.

► **(B) Pairwise deletion**

- Uses the observations in 1906-1910 when computing the means and covariances of emp and cpi.
- Uses the observations in 1906-1907 and 1909-1910 when computing the means and covariances of ip.
- Uses the observations in 1909-1910 when computing the means and covariances of gnp.
- Uses the observations in 1909-1910 for model estimation other than means and covariances.

► **(C) Variable deletion**

- Uses only emp and cpi for all parts of analysis.

►► 4.8 DATA IMPUTATION

- Imputing data replaces missing values with statistically determined values. Methods of imputation can vary from simply replacing missing values with the mean to sophisticated multiple imputation processes.
- Which method of imputation should be used depends on the characteristics of the data.

Table 4.8.1 : Examples of imputation methods

Method	Description	Advantages	Disadvantages
Replacement with mean, median, mode	All missing values are replaced with the variable mean, median or mode.	Easy to implement.	• Distorts the distribution of the data. • Reduces data variance. • Results in biased estimates.
Linear Regression	Missing values are predicted using a linear model and the other variables in the dataset	• Simple to implement. • Uses all available information.	• Biased correlations between variables. • Underestimated variability. • Falsely strengthens relationship between variables.
Last observation carried forward (LOCF)	• Used for time series data. • Use last observed data value as a replacement for missing data	• Appropriate for time series data. • Easy implementation.	• Can results in biased estimates even when data is MCAR. • Modeling techniques should address that data has been imputed by LOCF. • May incorrectly suggest stability across time stretches if used to fill successions of missing data.
Predictive mean matching	Replacements for missing values are drawn randomly from a group of nearby candidate values.	• Appropriate for time series data. • Easy to use and versatile. • Robust to data transformations. • Valid for discrete data. • Will always produce replacements within the observed data range.	• May duplicate values, especially if sample is small. • Not suited for small samples, skewed data, or sparse data. • Cannot be used to extrapolate beyond range of the data.

In statistics, **imputation** is the process of replacing **missing data** with substituted values. When substituting for a data point, it is known as “**unit imputation**”; when substituting for a component of a data point, it is known as “**item imputation**”.

4.9 DATA TRANSFORMATION

UQ. Explain the different modes of data transformation in big data. (SPPU - Q. 5(b). May 18, 8 Marks)

- Data transformation is the process of changing the format, structure, or values of data.
- For data analytics projects, data may be transformed at two stages of the data pipeline. Organizations that use on-premises data warehouses generally use an ETL (extract, transform, load) process, in which **data transformation is the middle step**.
- Today, most organizations use cloud-based data warehouses, which can scale compute and storage resources with latency measured in seconds or minutes. The scalability of the cloud platform lets organizations skip preload transformations and load raw data into the data warehouse, then transform it at query time - a model called ELT (**extract, load, transform**).
- Processes such as **data integration, data migration, data warehousing, and data wrangling** all may involve data transformation.
- Data transformation may be constructive (adding, copying, and replicating data), destructive (deleting fields and records), aesthetic (standardizing salutations or street names), or structural (renaming, moving, and combining columns in a database).
- An enterprise can choose among a variety of **ETL tools** that automate the process of data transformation. Data analysts, data engineers, and data scientists also transform data using **scripting languages such as Python or domain-specific languages like SQL**.

4.9.1 Benefits of Data Transformation

Transforming data yields several benefits :

- Data is transformed to make it better-organized. Transformed data may be easier for both humans and computers to use.
- Properly formatted and validated data improves data quality and protects applications from potential landmines such as null values, unexpected duplicates, incorrect indexing, and incompatible formats.

- Data transformation facilitates compatibility between applications, systems, and types of data. Data used for multiple purposes may need to be transformed in different ways

4.10 DATA STANDARDIZATION

- Data Standardization is a data processing workflow that converts the structure of disparate datasets into a **Common Data Format**. As part of the Data Preparation field, Data Standardization deals with the transformation of datasets after the data is pulled from source systems and before it's loaded into target systems. Because of that, Data Standardization can also be thought of as the **transformation rules** engine in Data Exchange operations.
- Data Standardization enables the data consumer to analyze and use data in a consistent manner. Typically, when data is created and stored in the source system, it's structured in a particular way that is often unknown to the data consumer.
- Moreover, datasets that might be semantically related may be stored and represented differently, thereby making it difficult for a data consumer to aggregate or compare the datasets.

Data Standardization Use Cases

There are two main use case categories in Data Standardization: Source-to-Target Mapping, and Complex Reconciliation. We typically divide the former into two sub-categories thereby arriving at three use cases :

- **Simple mapping from external sources** : This use case handles on-boarding data from systems that are external to the organization, and mapping its keys and values to an output schema.
- **Simple mapping from internal sources** : This use case involves handling internal datasets that are based on inconsistent definitions and transforming them into a single trustworthy data set for the entire organization.
- **Complex reconciliation** : This use case involves the creation of complex calculated metrics that provide their own semantics based on defined business logic.

4.11 HANDLING CATEGORICAL DATA WITH 2 AND MORE CATEGORIES

- Categorical data is simply information aggregated into groups rather than being in numeric formats, such as Gender, Sex or Education Level. They are present in almost all real-life datasets, yet the current algorithms still struggle to deal with them.
- Take, for instance, XGBoost or most SKlearn models. If you try and train them with categorical data, you'll immediately get an error.
- Currently, many resources advertise a wide variety of solutions that might seem to work at first, but are deeply wrong once thought through. This is especially true for non-ordinal categorical data, meaning that the classes are not ordered (As it might be for Good = 0, Better = 1, Best = 2). A bit of clarity is needed to distinguish the approaches that Data Scientists should use from those that simply make the models run.

What Not To Do

(I) Label Encoding

One of the simplest and most common solutions advertised to transform categorical variables is Label Encoding. It consists of substituting each group with a corresponding number and keeping such numbering consistent throughout the feature.

Categorical Feature	Label Encoding
United States	1
United States	1
France	2
Germany	3
United Kingdom	4
France	2

Example of Label Encoding

This solution makes the models run, and it is one of the most commonly used by aspiring Data Scientists. However, its simplicity comes with many issues.

(II) Distance and Order

Numbers hold relationships. For instance, four is twice two, and, when converting categories into numbers directly, these relationships are created despite not existing between the original categories. Looking at the example before, United Kingdom becomes twice France, and France plus United States equals Germany.

Well, that's not exactly right...

This is especially an issue for algorithms, such as K-Means, where a distance measure is calculated when running the model.

Solutions

(III) One-Hot Encoding

One-Hot Encoding is the most common, correct way to deal with non-ordinal categorical data. It consists of creating an additional feature for each group of the categorical feature and mark each observation belonging (Value = 1) or not (Value = 0) to that group.

United States	France	Germany	United Kingdom
1	0	0	0
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1
0	1	0	0

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(IV) Example of One-Hot Encoding

- This approach is able to encode categorical features properly, despite some minor drawbacks. Specifically, the presence of a high number of binary values is not ideal for distance-based algorithms, such as Clustering models.
- In addition, the high number of additionally generated features introduces the curse of dimensionality. This means that due to the now high dimensionality of the dataset, the dataset becomes much more sparse.
- In other words, in Machine Learning problems, you'd need at least a few samples per each feature combination. Increasing the number of features means that we might encounter cases of not having enough observations for each feature combination.

(IV) Target Encoding

A lesser known, but very effective way of handling categorical variables, is Target Encoding. It consists of substituting each group in a categorical feature with the average response in the target variable.

Country	Target Variable	Target Encoding
United States	1	0.40
Germany	0	0.50
United States	0	0.40
United States	1	0.40
France	1	0.67
Germany	1	0.50
United States	0	0.40
France	1	0.67
United States	0	0.40
France	0	0.67

Example of Target Encoding

The process to obtain the Target Encoding is relatively straightforward and it can be summarised as :

1. Group the data by category.
2. Calculate the average of the target variable per each group.
3. Assign the average to each observation belonging to that group

This can be achieved in a few lines of code:

```
encodings = data.groupby('Country')['Target
Variable'].mean().reset_index()data =
data.merge(encodings, how='left',
on='Country')data.drop('Country', axis=1,
inplace=True)
```

Alternatively, we can also use the category_encoders library to use the TargetEncoder functionality.

Target Encoding is a powerful solution also because it avoids generating a high number of features, as is the case for One-Hot Encoding, keeping the dimensionality of the dataset as the original one.

4.12 STATISTICAL AND GRAPHICAL ANALYSIS METHODS

UQ. Explain pie chart and scatter plot.

(SPPU - Q. 7(b), Dec. 19, 8 Marks)

Techniques based on Mathematics and Statistics

- **Descriptive Analysis :** Descriptive Analysis considers the historical data, Key Performance Indicators and describes the performance based on a chosen benchmark. It takes into account past trends and how they might influence future performance.
- **Dispersion Analysis :** Dispersion in the area onto which a data set is spread. This technique allows data analysts to determine the variability of the factors under study.
- **Regression Analysis :** This technique works by modeling the relationship between a dependent variable and one or more independent variables. A regression model can be linear, multiple, logistic, ridge, non-linear, life data, and more.
- **Factor Analysis :** This technique helps to determine if there exists any relationship between a set of variables. This process reveals other factors or variables that describe the patterns in the relationship among the original variables. Factor Analysis leaps forward into useful clustering and classification procedures.
- **Discriminant Analysis :** It is a classification technique in data mining. It identifies the different points on different groups based on variable measurements. In simple terms, it identifies what makes two groups different from one another; this helps to identify new items
- **Time Series Analysis:** In this kind of analysis, measurements are spanned across time, which gives us a collection of organized data known as time series.

Techniques based on Visualization and Graphs

- **Column Chart, Bar Chart :** Both these charts are used to present numerical differences between categories. The column chart takes to the height of the columns to reflect the differences. Axes interchange in the case of the bar chart.

- **Line Chart** : This chart represents the change of data over a continuous interval of time.
- **Area Chart** : This concept is based on the line chart. It also fills the area between the polyline and the axis with color, representing better trend information.
- **Pie Chart** : It is used to represent the proportion of different classifications. It is only suitable for only one series of data. However, it can be made multi-layered to represent the proportion of data in different categories.
- **Funnel Chart** : This chart represents the proportion of each stage and reflects the size of each module. It helps in comparing rankings.
- **Word Cloud Chart** : It is a visual representation of text data. It requires a large amount of data, and the degree of discrimination needs to be high for users to perceive the most prominent one. It is not a very accurate analytical technique.
- **Gantt Chart** : It shows the actual timing and the progress of the activity compared to the requirements.
- **Radar Chart** : It is used to compare multiple quantized charts. It represents which variables in the data have higher values and which have lower values. A radar chart is used for comparing classification and series along with proportional representation.
- **Scatter Plot** : It shows the distribution of variables in points over a rectangular coordinate system. The distribution in the data points can reveal the correlation between the variables.
- **Bubble Chart** : It is a variation of the scatter plot. Here, in addition to the x and y coordinates, the bubble area represents the 3rd value.
- **Gauge** : It is a kind of materialized chart. Here the scale represents the metric, and the pointer represents the dimension. It is a suitable technique to represent interval comparisons.
- **Frame Diagram** : It is a visual representation of a hierarchy in an inverted tree structure.
- **Rectangular Tree Diagram** : This technique is used to represent hierarchical relationships but at the same level. It makes efficient use of space and represents the proportion represented by each rectangular area.

• Map

- **Regional Map** : It uses color to represent value distribution over a map partition.
- **Point Map** : It represents the geographical distribution of data in points on a geographical background. When the points are the same in size, it becomes meaningless for single data, but if the points are as a bubble, it also represents the size of the data in each region.
- **Flow Map** : It represents the relationship between an inflow area and an outflow area. It represents a line connecting the geometric centers of gravity of the spatial elements. The use of dynamic flow lines helps reduce visual clutter.
- **Heat Map** : This represents the weight of each point in a geographic area. The color here represents the density.

4.13 HIVE DATA ANALYTICS

UQ. Explain Architecture of HIVE.

(SPPU - Q. 6(a), Dec. 19, 8 Marks)

GQ. What is Hive ?

- Hive, originally developed by Facebook and later owned by Apache, is a data storage system that was developed with a purpose to analyze organized data. Working under an open-source data platform called **Hadoop**, Apache Hive is an application system that was released in the year 2010 (October).
- Introduced to facilitate fault-tolerant analysis of hefty data on a regular basis, Hive has been used in **big data analytics** and has been popular in the realm for more than a decade now.
- Even though it has many competitors like Impala, Apache Hive stands apart from the rest of the systems due to its fault-tolerant nature in the process of data analysis and interpretation.

Understanding Hive in Big Data

- Apache Hive is a particularly efficient tool when it comes to big data (exponential data that is to be analyzed). A warehouse data software that supports the data analysis process of big data on a regular basis, the

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concept of hive big data is quite popular in the technological realm.

- As data is stored in the Apache Hadoop Distributed File System (HDFS) wherein data is organized and structured, Apache Hive helps in processing this data and analyzing it producing data-driven patterns and trends. Fit to be used by organizations or institutions, Apache Hive is extremely helpful in big data and its ever-changing growth.
- The concept of **Structured Query Language** or SQL software is involved in the process which communicates with numerous databases and collects the required data. Understanding Hive big data through the lens of data analytics can help us get more insights into the working of Apache Hive.
- By using a batch processing sequence, Hive generates data analytics in a much easier and organized form that also requires less time as compared to traditional tools. HiveQL is a language similar to SQL that interacts with the Hive database across various organizations and analyses necessary data in a structured format.

Why do we need it ?

- Hive in big data is a milestone innovation that has eventually led to data analysis on a large scale. Big

organizations need big data to record the information that is collected over the time.

- To produce data-driven analysis, organizations gather data and use such software applications to analyze their data. This data, with Apache Hive, can be used for reading, writing, and managing information that has been stored in an organized form. Ever since data analytics has come into being, storage of data has been a trending topic.
- Even though small scale organizations were able to manage medium-sized data and analyze it with traditional data analytics tools, big data could not be managed with such applications and so, there was a dire need for advanced software.
- As data collection became a daily task and organizations expanded in all aspects, data collection became exponential and vast. Furthermore, data began to be dealt in petabytes that define storage of vast data.
- For this, organizations needed hefty equipment and perhaps that is the reason why the release of a software like Apache Hive was necessary. Thus, Apache Hive was released with the purpose of analyzing big data and producing data-driven analogies.

Chapter Ends...

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